Kelvin Jonathan Yoga

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Skills

Post-Quantum Cryptography (PQC) · Zero-Knowledge Proofs · Zero-trust architecture · Crypto-Agility · Public-Key Infrastructure (PKI) · Networks Security · Classical-Quantum Tradeoffs · Resource Estimation · Linear Algebra · Optimization Algorithms · Python · C++ · Rust · Git · Docker · CI/CD Pipelines · Agile/Scrum.

Education

University of New South Wales – BS in Computer Science (Security Engineering) 2026

Experience

Applied Cryptographer – Clave Systems July 2026 - Present

- Collaborating in designing a secure authentication and data verification system in a zero trust environment
- Benchmarked existing tools and systems to see which fits best
- Implementing Cryptographic systems

PQ Cryptography Researcher – UNSW Apr 2026 - Present

- Leading a research project under Dr. Sushmita Ruj focused on privacy-preserving, scalable cryptographic protocols and credential migration frameworks.
- Designing, evaluating, and formally proving post-quantum group authentication protocols and dynamic ZK-proof systems on zero-trust architecture.
- Developing Proof of Concept (PoC) software implementations of the research on the core algorithms and zk-proof integrations, benchmarking performance efficiency, proof sizes.

Security & Backend Engineer – Solar Central March 2026 - May 2026

- Developed secure Python backends, SQL data access objects, and input validation engines within an Agile team of 6 utilizing Scrum workflows.
- Conducted code-based security audits and technical briefings to communicate system vulnerability risks and map out mitigation roadmaps.
- Built and maintained automated, reproducible testing and CI/CD pipelines to ensure rapid, secure deployment and system resilience.

IT Support & Data Entry – Sydney Endocrinology and Diabetes September 2023 - December 2024

- Managed office hardware, secure local network infrastructure, and automated encrypted data backups to maintain system reliability.
- Maintained 100% accuracy in data handling while strictly adhering to data privacy standards, access controls, and compliance frameworks.

Investigations

Reed-Solomon Error-Correcting Code (ongoing)

- Engineering a complete Reed-Solomon encoder and decoder in C++ utilizing Galois Field arithmetic.
- Applying coding theory principles and polynomial operations to implement robust data integrity and fault tolerance.
- Developing custom polynomial division and multiplication algorithms to optimize encoding throughput.

C-Based Proxy Server with Request Filtration

- Implemented a full-scale network proxy server in C from scratch using socket programming, incorporating request filtration, caching strategies, and HTTPS tunneling.
- Managed low-level network performance variables including response timeouts, socket re-use, and full HTTP/HTTPS protocol compliance.

DIMY Privacy-Preserving Protocol Implementation

- Implemented the DIMY privacy-centric contact tracing protocol, engineering a custom socket-based network layer (TCP/UDP) for concurrent, low-latency data exchange.
- Executed adversarial threat modeling assessments to isolate architectural flaws, designing a target cryptographic validation mechanism to ensure long-term system integrity.

Multi-Protocol Fuzzing Engine

- Developed a custom mutation-based fuzzing script engine in Python to systematically analyze software input data validation patterns and isolate system dependencies.
- Authored technical assessment reports and clear documentation outlining systemic bugs found within multi-format data schemas.

Post-Quantum Digital Signature Scheme System PoC

- Researched and rewrote a researcher's aggregate signature scheme PoC from Rust into C++ in full, within a three-person team.
- Investigated feasibility of integrating STARK/SNARK proof systems(libiop, starkware) into the aggregation layer.
- Conducted peer review on an alternative Lattice-based scheme proposed solution.

Technical Training

OffSec Wireless Professional (OSWP) Curriculum – Syllabus Completion 2026
Comprehensive self-study of the syllabus with a defensive mindset; focused heavily on enterprise authentication mechanisms, EAP protocol handshakes, and network-layer security validation
IBM Quantum Learning – in progress
Self-study on Quantum Algorithms with the IBM Quantum Computing course on Quantum computation.

Achievements

AI X Creativity Hackathon 1st Place – Awarded 1st place out of 15 competing technical teams at the University of Sydney.